

COMS20018, Introduction to AI:

Unit Introduction

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21st September 2026

Welcome

- ▶ Welcome to COMS20018, Introduction to Artificial Intelligence!
- ▶ Lecturers:



Mengyan Zhang
Unit Director



Nan Lu



James Cussens

- ▶ Teaching Assistants:
 - ▶ Yuming Li, Siddhant Bansal

Where to Find Everything

- ▶ Unit GitHub page (public):
 - ▶ <https://uob-coms20018.github.io/>
 - ▶ Schedule, lecture slides, and lab notebooks, updated as the unit progresses
- ▶ Blackboard Ultra page:
 - ▶ Used for material that can't be public, e.g. assessment submission
 - ▶ Unit announcements
- ▶ Teams chat:
 - ▶ Ask questions

What This Unit Covers

- ▶ The foundations and contemporary landscape of Artificial Intelligence:
 - ▶ history, theory, practical applications, and philosophy
- ▶ Core topics:
 - ▶ Machine learning (regression/classification, neural networks, generative models)
 - ▶ Large language models
 - ▶ Causal inference · Reinforcement learning · Computer vision
 - ▶ Philosophy of AI · Ethics · Trustworthy AI
- ▶ Reading materials are listed on the unit website
- ▶ A hands-on approach throughout, via weekly labs.

What Background You Need

- ▶ This unit builds directly on first-year COMS foundations. You should already have:

Programming

COMS10016
Imperative & Functional
+ COMS10018
OOP & Algorithms

Computer Systems

COMS10015
Computer Architecture
+ COMS10012
Software Tools

Mathematics

COMS10014
Maths for CS A
+ COMS10013
Maths for CS B

- ▶ In short: comfort with programming, algorithmic thinking, and discrete maths / mathematical foundations for computing.
- ▶ All coding on this unit is in Python; we will cover Introduction to Python in the first week's lab.

Learning Outcomes

- ▶ By the end of this unit, you will be able to:
 - ▶ Identify the opportunities and challenges of different AI approaches
 - ▶ Deploy, interact with, and evaluate AI systems
 - ▶ Explain the theoretical insights underpinning AI approaches
 - ▶ Discuss how to apply AI methods to novel problems, effectively and responsibly

Unit Schedule

Lecture 1: Mondays, 1100-1200, Fry Building G.09

Lecture 2: Tuesdays, 1500-1600, Fry Building G.09

Lab/Seminar: Wednesdays, 1100-1300, Merchant Venturer's Building 1.15 (Linux Computer Lab)

Week	Topic	Lecturer	Lecture 1 (Mon, 1100-1200)	Lecture 2 (Tue, 1500-1600)	Lab/Seminar (Wed, 1100-1300)
1 (w/c 21/09/26)	Introduction	MZ	Unit introduction; Introduction to AI	Linear Regression and Classification: Part I	Lab:Introduction to Python
2 (w/c 28/09/26)	Machine Learning	MZ	Linear Regression and Classification: Part II	Neural Networks: Introduction,	Lab: Linear Regression
3 (w/c 05/10/26)	Deep Learning	NL / MZ	Neural Networks: CNNs, RNNs, and Transformers (MZ)	Generative Models	Lab: Neural Networks
4 (w/c 12/10/26)	Large Language Models	NL	Large Language Models	Large Language Models	Lab: Large Language Models
5 (w/c 19/10/26)	Philosophy of AI	JC	Philosophy of AI	Philosophy of AI	Seminar
6 (w/c 26/10/26)	Consolidation week - no scheduled teaching				
7 (w/c 02/11/26)	Ethics	NL	Ethics	Ethics	Lab
8 (w/c 09/11/26)	Causality and Decision Making	MZ	Causal Inference	Sequential Decision Making	Lab: Causality and Decision Making
9 (w/c 16/11/26)	Reinforcement Learning	MZ	Reinforcement Learning	Reinforcement Learning	Lab: Reinforcement Learning
10 (w/c 23/11/26)	Computer Vision	NL / Guest	Computer Vision	Computer Vision (Guest lecture: Ahmad Darkhalil)	Lab
11 (w/c 30/11/26)	Trustworthy AI	NL	Trustworthy AI	Trustworthy AI	Lab
12 (w/c 07/12/26)	Revision				No lab
13 (w/c 14/12/26)	Final Exam period				

Full schedule, kept up to date, at uob-coms20018.github.io.

Teaching Sessions

- ▶ Lecture 1: Mondays, 11:00–12:00, Fry Building G.09
- ▶ Lecture 2: Tuesdays, 15:00–16:00, Fry Building G.09
- ▶ Lab/Seminar: Wednesdays, 11:00–13:00, MVB 1.15 (Linux computer lab)
- ▶ Week 6 is a consolidation week, no scheduled teaching.
- ▶ Week 12 is revision week (no lab, first lecture as a drop-in session).

Assessment

- ▶ You are assessed via three elements:

20%

Mid-term video
presentation

20%

Lab-based
assessments

60%

End-of-term exam

- ▶ A mid-term video presentation (20% of the unit)
- ▶ A series of laboratory-based assessments (20% of the unit)
- ▶ An end-of-term exam (60% of the unit), closed book

Assessment: Mid-term Video (20%)

- ▶ **Format:** individual, up to 15 minutes, submitted as a video file on Blackboard.
- ▶ **Task:** choose one topic → one question → 1–2 papers to discuss in relation to it.
- ▶ **Scope:** topics drawn from Weeks 2–5 (Machine Learning → Deep Learning → LLMs → Philosophy of AI).
- ▶ **Timing:** topic/question/paper menu will be released by the end of Week 3; deadline is mid Week 7.
- ▶ Full topic list, marking criteria, and submission details will be posted on Blackboard.

Assessment: Lab-Based Work (20%)

- ▶ **Weekly cycle:** notebook released Wednesday → you submit your completed notebook by **Sunday 23:59** → answers released the following Monday.
- ▶ **Assessed weeks:** 3, 4, 7, 8, 9, 11
- ▶ **Best-5-of-6:** your 1 lowest-scoring lab is automatically dropped, so the 20% is earned from your best 5 labs, each worth 4% of the unit.
- ▶ Full submission and marking details will be posted on Blackboard.

Labs & Software Setup

- ▶ All coding is in Python, using packages such as `scikit-learn` and `PyTorch`.
- ▶ We use Jupyter Notebook, running in Google Colab; you can also use:
 - ▶ a UoB machine (MVB 1.15):
 - ▶ Reboot into Linux if necessary
 - ▶ Open a terminal and run: `module load anaconda/3-2025`
 - ▶ your own machine:
 - ▶ Install [Anaconda](https://anaconda.com) (anaconda.com)
 - ▶ We're happy to help with install problems, but can't guarantee to fix everything

Getting Help

- ▶ Feel free to ask questions during labs and seminars.
- ▶ Post questions on Teams chat.
- ▶ Full details (assessment timing, submission, format) will be posted on Blackboard as they are confirmed.

Any questions?